

## **Professional Summary**

- AI/ML Engineer with 4+ years of experience in designing, deploying, and scaling end-to-end machine learning solutions across NLP, Computer Vision, Generative AI (LLMs), and advanced analytics using deep learning and statistical modeling.
- Proficient in building end-to-end ML pipelines using Python, Scikit-learn, Pandas, and MLflow well-versed in feature engineering, model evaluation, hyperparameter tuning, and deploying models in containerized environments (Docker, FastAPI, Flask).
- Expert in fine-tuning deep learning models using PyTorch, TensorFlow, and Keras including CNNs for image classification, LSTMs for time series forecasting and sentiment analysis optimized for latency and throughput on cloud-based GPU/TPU instances.
- Experienced in transformer-based architectures (BERT, LLaMA2, GPT-4) and RAG pipelines using LangChain, Hugging Face, and vector databases (Pinecone, FAISS) for semantic search, document understanding, and conversational AI.
- Skilled in ML infrastructure management and deployment using AWS (SageMaker, EC2, S3, Lambda), Google Cloud (Vertex AI, BigQuery), Azure DevOps, Docker, CI/CD pipelines and MLOps best practices for production-grade AI systems.
- Strong background in statistical analysis, A/B testing, and data visualization, utilizing SQL, Python, Tableau, and Power BI to extract business insights from large datasets and drive data-informed decision-making in AI-driven projects.

## **Education**

**Master of Science in Computer Science (GPA: 3.8/4) | Florida Atlantic University, Boca Raton, Florida**

## **Skills**

**Language/ IDE's:** Python, MATLAB, Jupyter Notebook, Google Colab, VS Code, SSMS

**Machine Learning:** Linear, Logistic Regression, Decision Trees, Random Forests, NumPy, SVM, A/B Testing, Naive Bayes

**Deep Learning:** CNN, RNN, LSTM, NLP, Large Language Model (LLM), Hugging Face Transformers (BERT, GPT-3)

**AI agent Frameworks:** LangChain, OpenAI agents SDK, AutoGen, Llama index

**Cloud/Visualizations:** AWS (SageMaker, EC2, SQS, SNS, Code Deploy, CloudWatch, API Gateway), GCP (Vertex AI, Google Cloud Storage), Tableau, Power BI

**Statistical Techniques:** Hypothesis Testing, Data Visualization, Data Modelling, A/B testing, Model Evaluation

**Packages and Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn, TensorFlow, Keras, NLTK, XGBoost, PyTorch

**Database and Tools:** SQL Server, MySQL, PostgreSQL, Redis

**Certifications:** Microsoft Certified: Azure AI Engineer Associate, AWS Cloud Quest: Cloud Practitioner, Microsoft Certified: Azure Fundamental

## **Work Experience**

### **Morgan Stanley, Boca Raton, FL**

**Nov 2024 – Present**

#### **Machine Learning Engineer**

- Built a high-impact credit risk prediction model using XGBoost and financial attributes to assess borrower profiles, which improved loan approval accuracy by 32% and minimized default rates in underserved financial segments.
- Automated machine learning pipelines on AWS using SageMaker, Glue, Lambda and S3 using PyTorch and TensorFlow and reducing data engineering overhead by 60%
- Designed an advanced anomaly detection pipeline using Isolation Forest on historical claims and trading data to detect unusual patterns, which enabled fraud analysts to reduce investigation time by 45% and comply with federal audit standards.
- Built custom financial NLP pipelines using spaCy and regex-based matchers to extract financial tickers, fund classifications, and macroeconomic indicators, enhancing report parsing speed by 39% and supporting faster insights for research analysts.
- Created Power BI dashboards to continuously monitor key model performance metrics including inference speed, attention drift, and threshold sensitivity, providing actionable insights for the product
- Automated compliance documentation and regulatory report generation via Python scripting and scheduling tools, cutting manual report preparation by 70%, improving audit-readiness and maintaining alignment with SEC and FINRA standards.
- Led A/B testing on alternative modeling strategies such as Random Forest and Gradient Boosting for financial behaviour predictions, which uncovered a superior model pipeline that improved model precision by 21%.

### **Accenture, Coimbatore, India**

**Aug 2021 – Jul 2023**

#### **AI & Deep Learning Developer**

- Architected cutting-edge deep learning models for financial time-series forecasting, significantly improving asset price prediction accuracy by 22%, enhancing algorithmic trading strategies and portfolio optimization
- Implemented NLP pipelines using BERT transformers to analyse financial news and social media sentiment, enhancing trading signal precision by 18% and improving market responsiveness
- Integrated OpenAI GPT-3 with prompt engineering and few-shot learning to generate real-time summaries of lengthy customer logs. This initiative reduced average response time by 41% and significantly enhanced customer support performance.
- Executed scalable, cloud-native machine learning workflows on Google Cloud Platform BigQuery and Vertex AI, reducing model retraining time and risk evaluation latency by 50% while ensuring high availability
- Applied unsupervised learning and autoencoder techniques for transaction anomaly detection, reducing false positives by 30% and strengthening fraud detection capabilities
- Streamlined end-to-end CI/CD pipelines for machine learning lifecycle management with Docker, Kubernetes, and GitHub Actions, reducing deployment times by 40% and ensuring seamless production rollouts
- Partnered with quantitative analysts, traders, and compliance teams to ensure models adhered to regulatory standards and business KPIs, improving transparency, reliability, and stakeholder trust

### **Accenture, Coimbatore, India**

**Jun 2020–Jul 2021**

#### **AI/ML Engineer**

- Devised deep learning models including convolutional and recurrent neural networks for automated document classification and object detection, increasing operational throughput by 40% and accelerating enterprise AI adoption
- Optimized and fine-tuned NLP models using BERT using Hugging Face Transformers for sentiment analysis and intent detection, improving classification accuracy by 20% and reducing customer escalation times
- Formed reusable PyTorch and TensorFlow pipelines with automated hyperparameter tuning and batch inference, reducing model training time by 60% and ensuring reproducibility and scalability across deployments

- Formulated and deployed real-time AI services using OpenCV and LSTM networks to detect abnormal behaviors in video streams, reducing manual fraud monitoring efforts by 55% and strengthening compliance controls
- Transformed legacy machine learning models to AWS SageMaker using containerized deployments and established API Gateway and Lambda endpoints, reducing deployment time by 50% and enabling seamless model scaling
- Orchestrated model evaluation, logging, and drift detection with MLflow and AWS CloudWatch, enhancing governance, reducing production incidents by 40%, and ensuring continuous model performance
- Crafted and maintained Tableau dashboards integrated with prediction confidence intervals and model impact KPIs, boosting stakeholder decision-making efficiency by 30% and accelerating data-driven strategy adoption across departments